References

[1] Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q. V., and Zhou, D. (2022). Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. NeurIPS 35. https://arxiv.org/abs/2201.11903

[2] Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., and Schulman, J. (2021). Training Verifiers to Solve Math Word Problems. https://arxiv.org/abs/2110.14168

[3] Hendrycks, D., Burns, C., Kadavath, S., Arora, A., Basart, S., Tang, E., Song, D., and Steinhardt, J. (2021). Measuring Mathematical Problem Solving With the MATH Dataset. Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks, 1. https://datasets-benchmarks-proceedings.neurips.cc/paper_files/paper/2021/hash/be83ab3ecd0db773eb2dc1b0a17836a1-Abstract-round2.html

[4] Clark, P., Cowhey, I., Etzioni, O., Khot, T., Sabharwal, A., Schoenick, C., and Tafjord, O. (2018). Think You Have Solved Question Answering? Try ARC, the AI2 Reasoning Challenge. https://arxiv.org/abs/1803.05457

[5] Rein, D., Hou, B. L., Stickland, A. C., Petty, J., Pang, R. Y., Dirani, J., Michael, J., and Bowman, S. R. (2023). GPQA: A Graduate-Level Google-Proof Q&A Benchmark. COLM 2024. https://arxiv.org/abs/2311.12022

[6] Lightman, H., Kosaraju, V., Burda, Y., Edwards, H., Baker, B., Lee, T., Leike, J., Schulman, J., Sutskever, I., and Cobbe, K. (2023). Let's Verify Step by Step. ICLR 2024. https://arxiv.org/abs/2305.20050

[7] Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., and Zhou, D. (2022). Self-Consistency Improves Chain of Thought Reasoning in Language Models. ICLR 2023. https://arxiv.org/abs/2203.11171

[8] Chen, X., Xu, J., Liang, T., He, Z., Pang, J., Yu, D., Song, L., Liu, Q., Zhou, M., Zhang, Z., Wang, R., Tu, Z., Mi, H., and Yu, D. (2024). Do NOT Think That Much for 2+3=? On the Overthinking of o1-Like LLMs. https://arxiv.org/abs/2412.21187

[9] Cho, K., van Merrienboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., and Bengio, Y. (2014). Learning Phrase Representations Using RNN Encoder-Decoder for Statistical Machine Translation. EMNLP, 1724-1734. https://aclanthology.org/D14-1179/

[10] Su, J., Lu, Y., Pan, S., Murtadha, A., Wen, B., and Liu, Y. (2021). RoFormer: Enhanced Transformer with Rotary Position Embedding. https://arxiv.org/abs/2104.09864

[11] Ferguson, T. S. Optimal Stopping and Applications. UCLA electronic text. Chapters 3 and 5. https://www.math.ucla.edu/~tom/Stopping/sr3.pdf and https://www.math.ucla.edu/~tom/Stopping/sr5.pdf

[12] Peskir, G., and Shiryaev, A. (2006). Optimal Stopping and Free-Boundary Problems. Birkhauser, Lectures in Mathematics ETH Zurich. https://doi.org/10.1007/978-3-7643-7390-0

[13] Hoeffding, W. (1963). Probability Inequalities for Sums of Bounded Random Variables. Journal of the American Statistical Association 58(301), 13-30. https://doi.org/10.1080/01621459.1963.10500830

[14] Howard, S. R., Ramdas, A., McAuliffe, J., and Sekhon, J. (2021). Time-uniform, nonparametric, nonasymptotic confidence sequences. Annals of Statistics 49(2), 1055-1080. https://doi.org/10.1214/20-AOS1991

[15] Clopper, C. J., and Pearson, E. S. (1934). The Use of Confidence or Fiducial Limits Illustrated in the Case of the Binomial. Biometrika 26(4), 404-413. https://doi.org/10.1093/biomet/26.4.404

[16] Graves, A. (2016). Adaptive Computation Time for Recurrent Neural Networks. arXiv:1603.08983. https://arxiv.org/abs/1603.08983

[17] Banino, A., Balaguer, J., and Blundell, C. (2021). PonderNet: Learning to Ponder. 8th ICML Workshop on Automated Machine Learning; arXiv:2107.05407. https://arxiv.org/abs/2107.05407

[18] Schuster, T., Fisch, A., Gupta, J., Dehghani, M., Bahri, D., Tran, V. Q., Tay, Y., and Metzler, D. (2022). Confident Adaptive Language Modeling. Advances in Neural Information Processing Systems 35. https://papers.neurips.cc/paper_files/paper/2022/hash/6fac9e316a4ae75ea244ddcef1982c71-Abstract-Conference.html

[19] Aggarwal, P., Madaan, A., Yang, Y., and Mausam. (2023). Let's Sample Step by Step: Adaptive-Consistency for Efficient Reasoning and Coding with LLMs. Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, 12375-12396. https://aclanthology.org/2023.emnlp-main.761/

[20] Liu, X., and Wang, L. (2025). Answer Convergence as a Signal for Early Stopping in Reasoning. Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, 17896-17907. https://aclanthology.org/2025.emnlp-main.904/

[21] Sun, R., Cheng, W., Li, D., Chen, H., and Wang, W. (2026). Stop When Enough: Adaptive Early-Stopping for Chain-of-Thought Reasoning. Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 27250-27268. https://aclanthology.org/2026.acl-long.1256/

[22] Ehab, M., El Gadarri, A., Farias, V. F., Jozefiak, A., and Moallemi, C. C. (2026). OS-Pruner: Pruning Chains-of-Thought of Reasoning Models via Optimal Stopping. arXiv:2607.11089, preprint. https://arxiv.org/abs/2607.11089

[23] Platt, J. C. (2000). Probabilities for SV Machines. In A. J. Smola, P. Bartlett, B. Schölkopf, and D. Schuurmans (eds.), Advances in Large-Margin Classifiers, 61-74. MIT Press. https://doi.org/10.7551/mitpress/1113.003.0008

[24] Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. (2017). On Calibration of Modern Neural Networks. Proceedings of the 34th International Conference on Machine Learning, Proceedings of Machine Learning Research 70, 1321-1330. https://proceedings.mlr.press/v70/guo17a.html

[25] Qwen Team. (2024). Qwen2.5 Technical Report. arXiv:2412.15115, first submitted December 19, 2024; revised January 3, 2025. https://arxiv.org/abs/2412.15115. Official Qwen2.5-0.5B-Instruct model card (accessed October 4, 2026): https://huggingface.co/Qwen/Qwen2.5-0.5B-Instruct